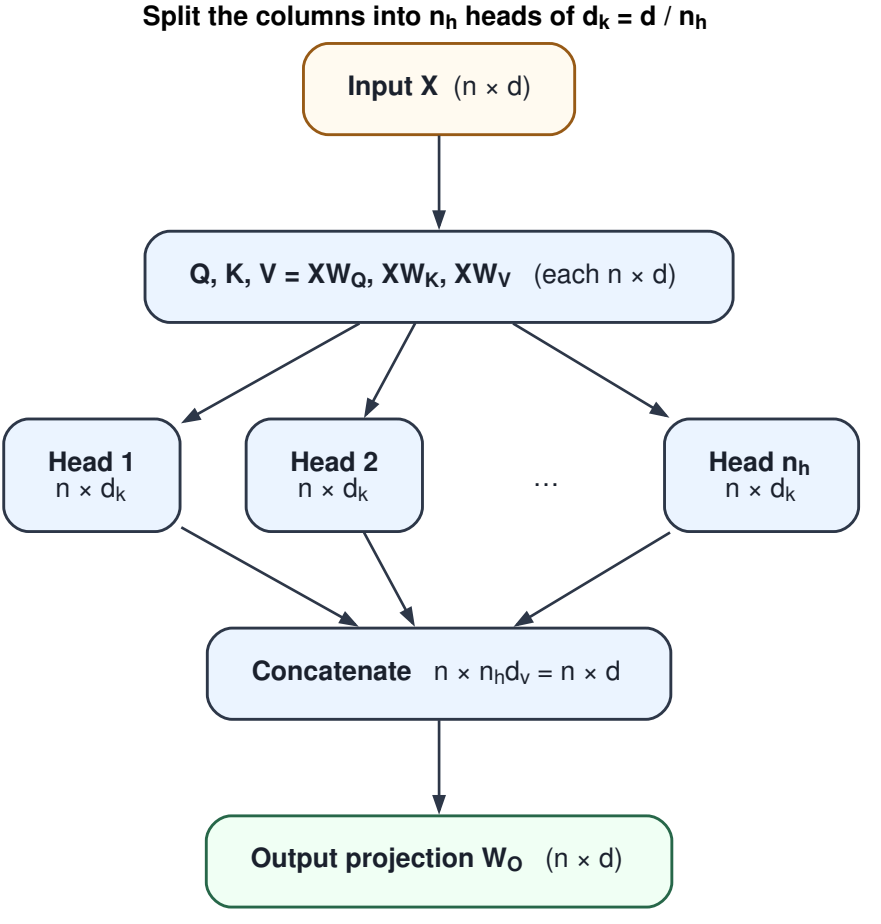
Split the columns into n_h heads of $d_k = d / n_h$
Input X (n × d)
Q, K, V = XW_Q, XW_K, XW_V (each n × d)
Head 1
n × d_k
Head 2
n × d_k
...
Head n_h
n × d_k
Concatenate n × $n_h d_v$ = n × d
Output projection W_O (n × d)